

Module 9: AI Agent Observability and AgentOPs

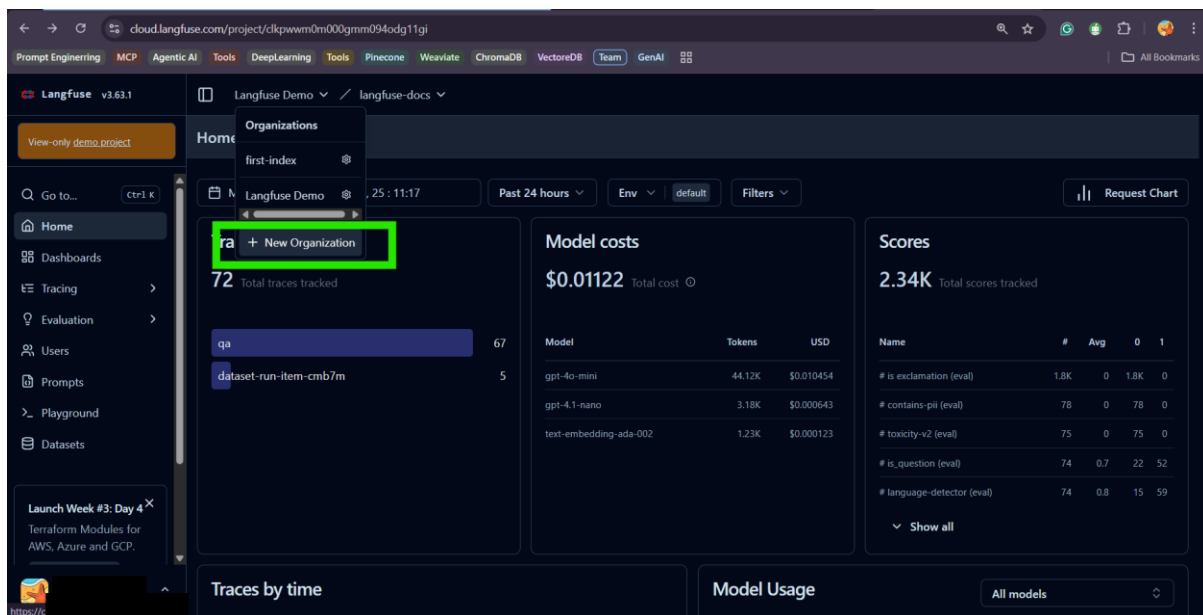
Demo-3

Demo-1: AI Observability with Langfuse

Langfuse: <https://cloud.langfuse.com/>

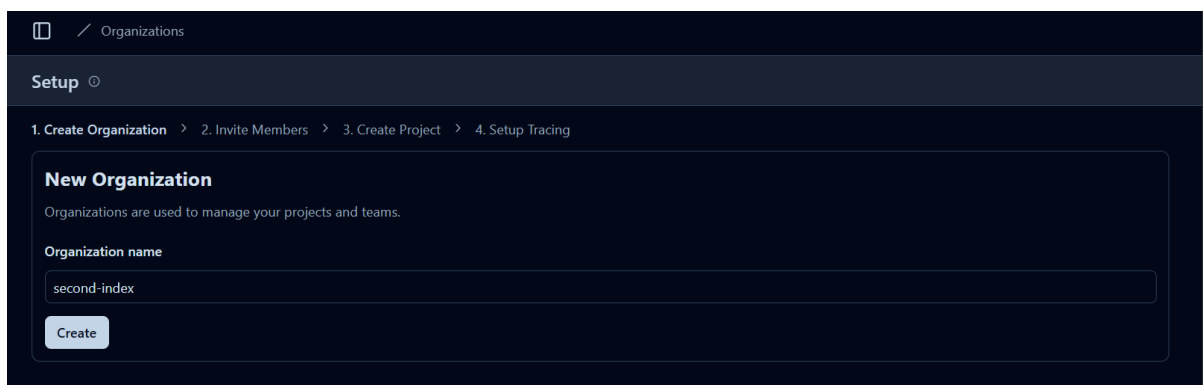
Langfuse Account Set-up

- Sign-up or log-in to your Langfuse Account.
- Once in, Click on “New Organization” to create one.

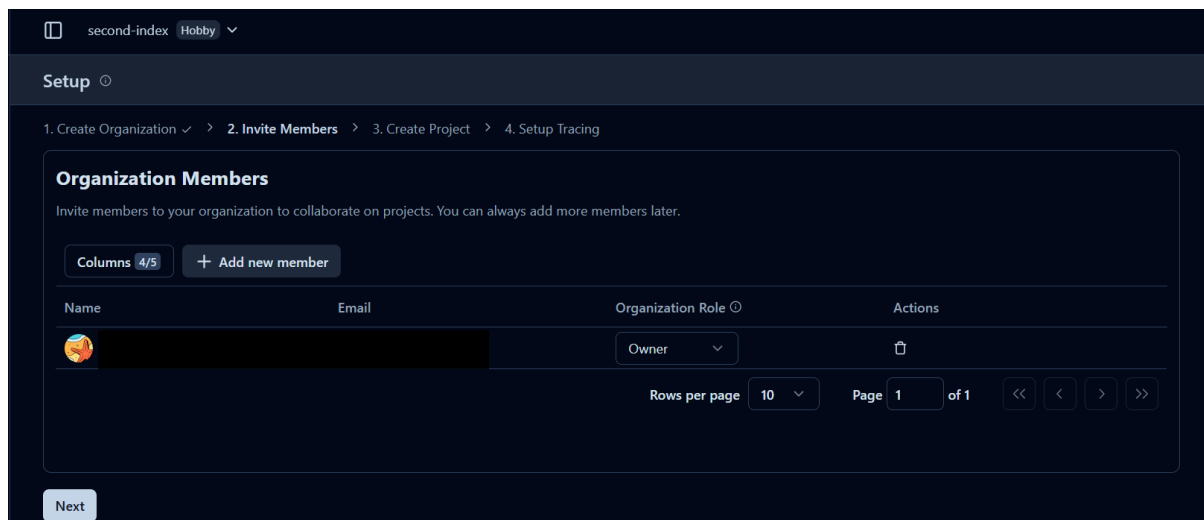


Give a name for your Organization

- Although in the screenshot of the Langfuse interface you can see the name as second-index, you can name your organization anything.
- Going forward you will notice that the name of the organization in the demo is “first-index”

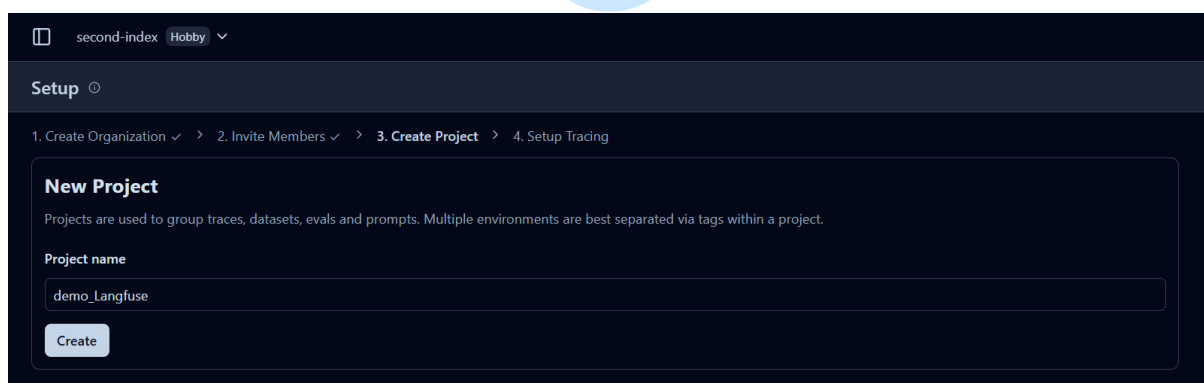


Organization Members

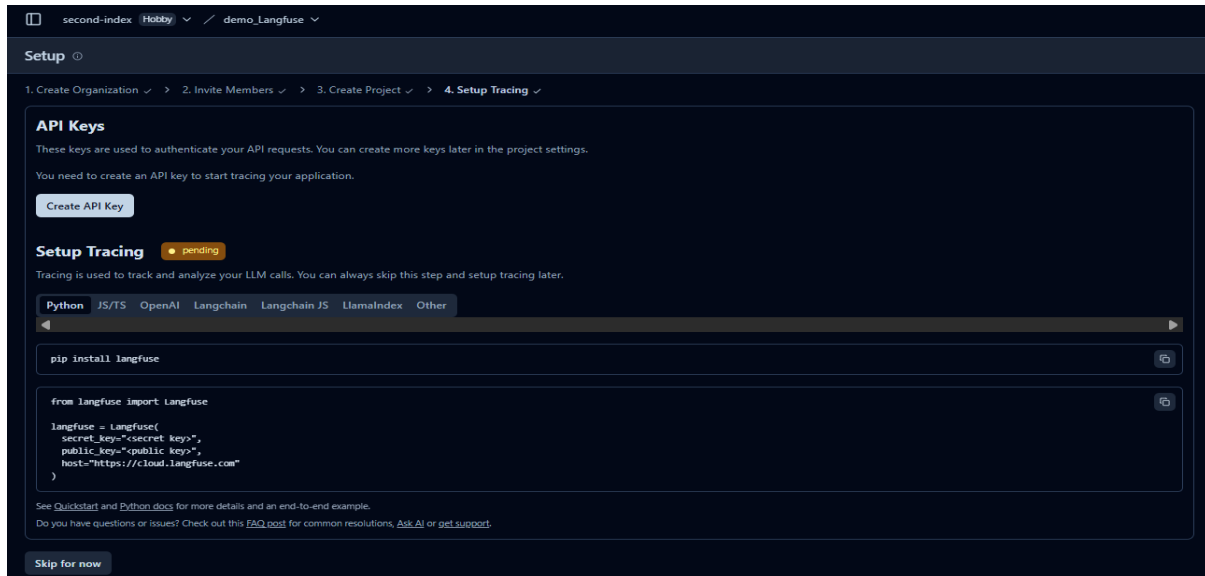


New Project: Give a name to your project just like you created the organization name.

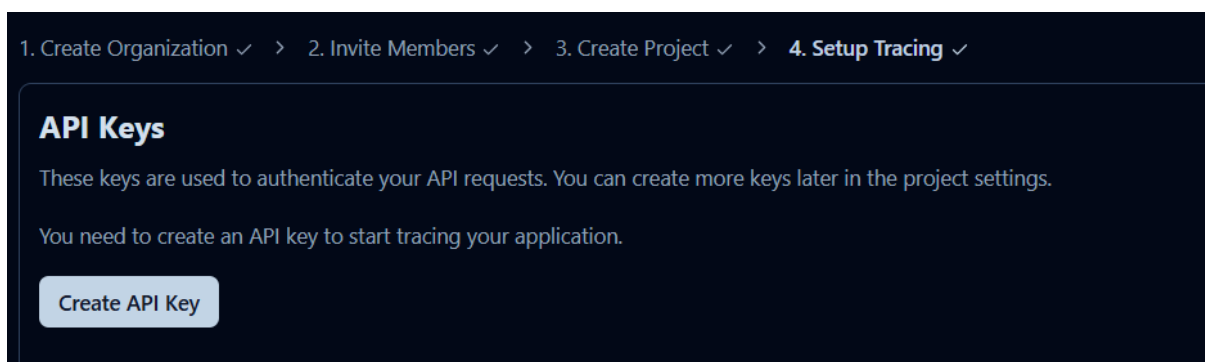
- Although in the screenshot of the Langfuse interface you can see the name as demo_Langfuse, you can name your project based on your preference.
- Going forward you will notice that the name of the project in the demo is “Langfuse_demo”



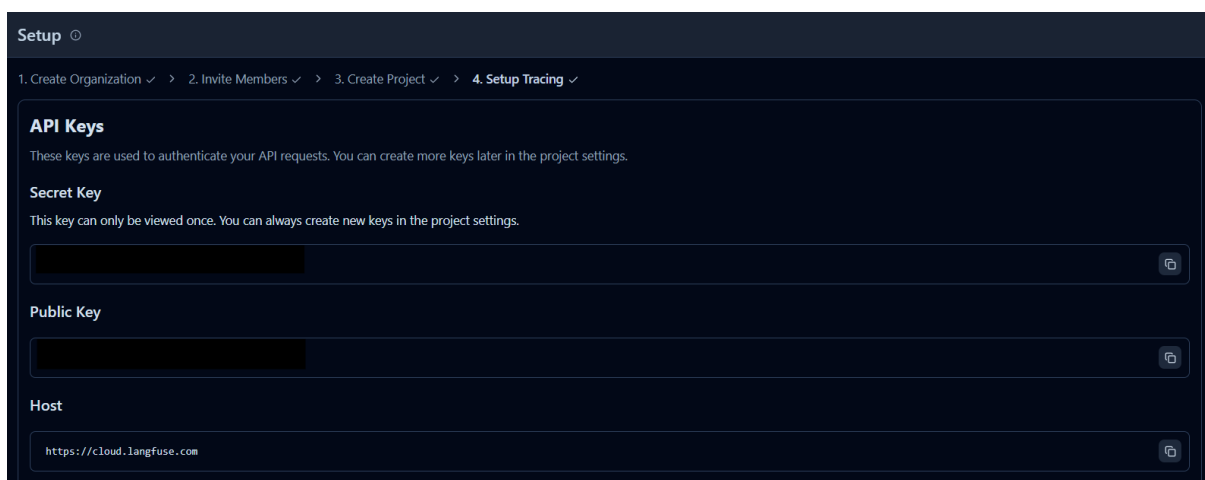
Setup Tracing



Create API Keys

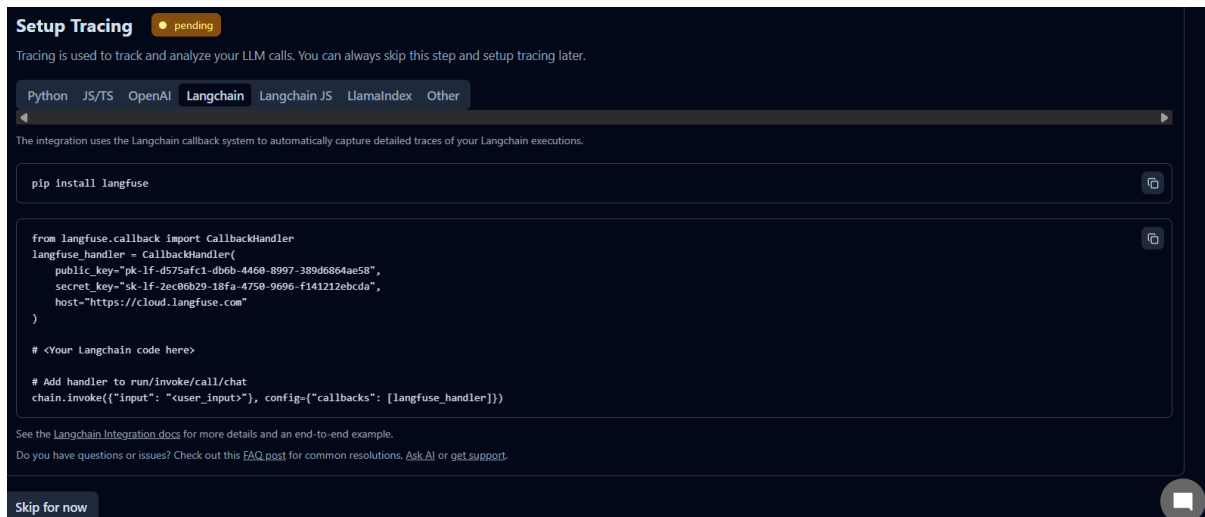


Store your API Keys safely



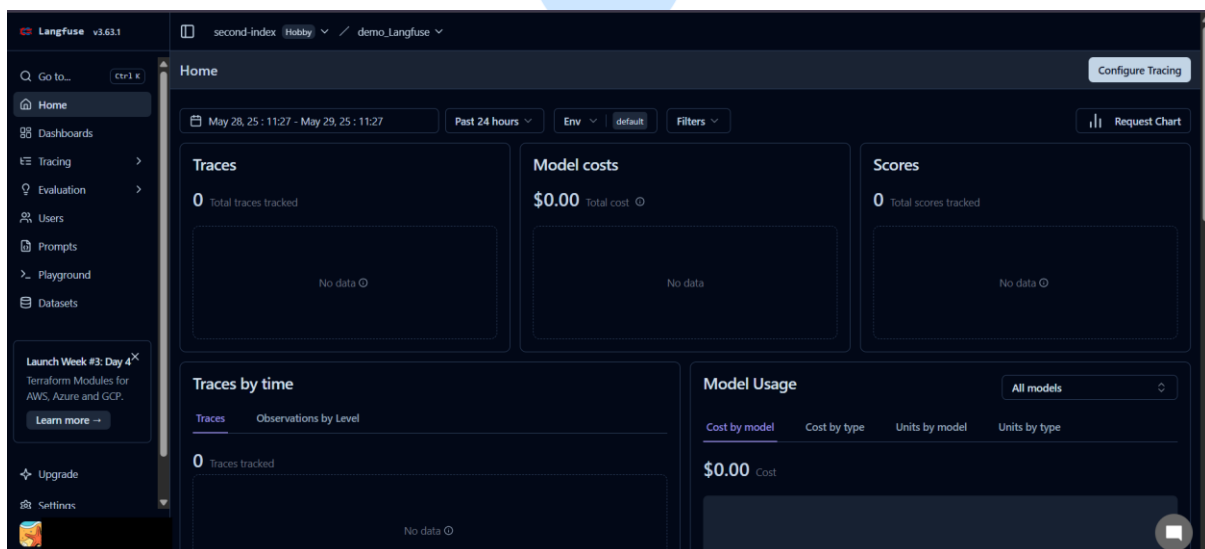
You will notice the “pending” indicator. Once you set-up your tracing in the application you want to trace, it turns to “active” and green colour indicator.

You can keep the code for guidance and then skip this step for now.



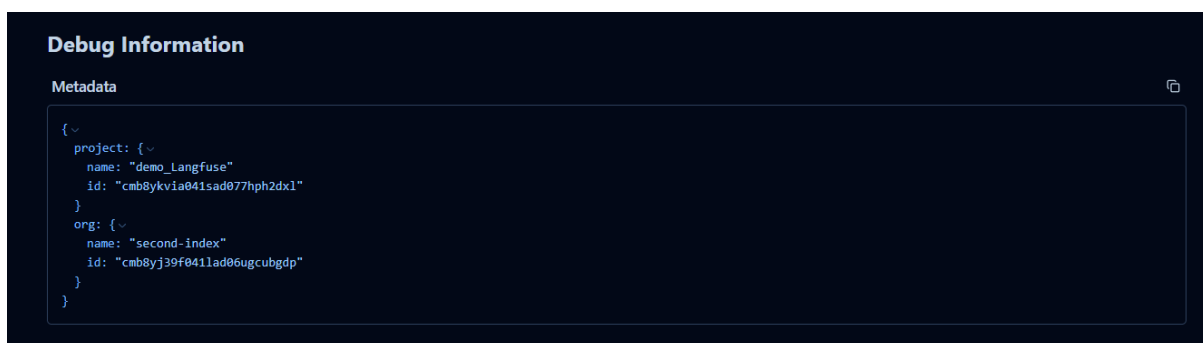
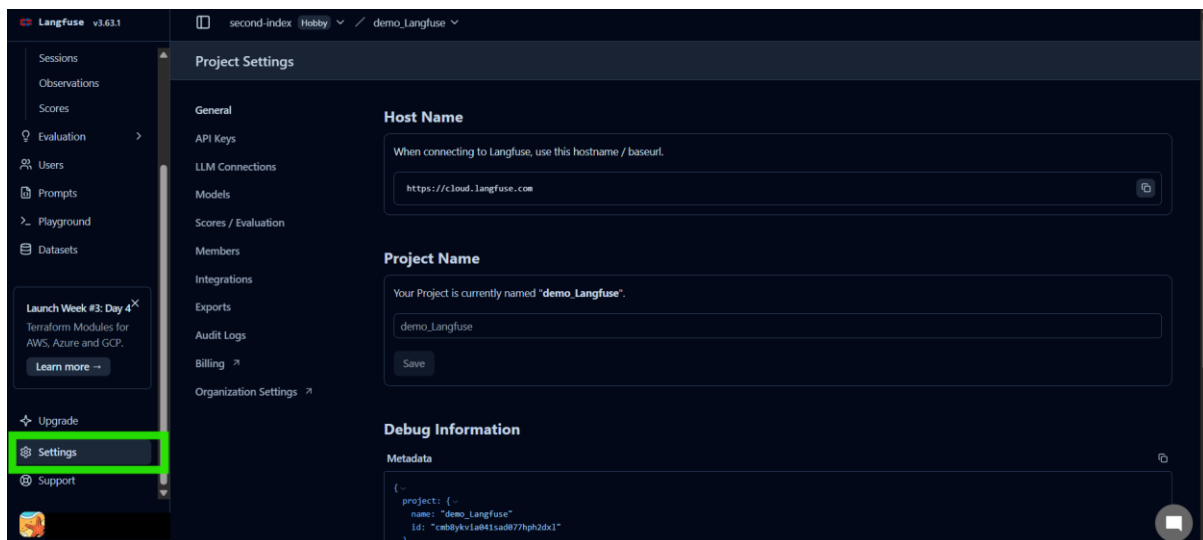
Project Setup complete

Once you are done setting up the project the Home tab will look like this.



Settings

You can view the different Project Settings here.



Overview

This demonstration showcases how to implement AI Observability using Langfuse in a multi-agent research application. The application leverages LangGraph to orchestrate a workflow of agents that collaboratively research a user-provided topic. The agents include a Search Agent (using DuckDuckGo), an Analyzer Agent, and a Summarizer Agent (both powered by the Groq API's llama-3.3-70b-versatile model). Langfuse is integrated to provide observability, enabling tracing, monitoring, and evaluation of the agents' performance. This setup ensures transparency into the AI system's behavior, making it easier to debug, evaluate, and improve the application. The demo runs in Google Colab and includes dynamic user input for research topics.

Tasks Achieved

- Multi-Agent Workflow: Built a LangGraph-based workflow with three agents:
 - Search Agent: Fetches recent developments on a topic using DuckDuckGo.
 - Analyzer Agent: Analyzes search results to identify key points and trends.

- Summarizer Agent: Summarizes the analysis into a concise research report.
- AI Observability with Langfuse:
 - Traced the execution of each agent, capturing inputs, outputs, and metadata.
 - Monitored token usage for Groq API calls.
 - Evaluated the final summary with a static score and comment.
- Dynamic Input: Allowed users to input a research topic interactively in Colab.
- Error Handling: Included basic error handling for API key retrieval, Groq API calls, and DuckDuckGo searches.

Project Structure

The project is a single Jupyter Notebook (Langfuse_demo.ipynb) with the following structure:

- Setup: Installs dependencies and imports libraries.
- API Key Retrieval: Retrieves API keys from Colab Secrets.
- Agent Definitions: Defines the Search, Analyzer, and Summarizer agents using LangGraph.
- Workflow Compilation: Sets up the LangGraph workflow with Langfuse tracing.
- Main Function: Executes the workflow, processes user input, and logs results with Langfuse.
- Observability Integration: Uses Langfuse to trace and evaluate the workflow.

PROJECT EXPLANATION

Setup

To run this demo, follow these steps to set up your environment and configure Langfuse for observability.

Step 1: Create a Google Colab Notebook:

- Open Google Colab and create a new notebook.
- Copy the code from the provided `Langfuse_demo.ipynb`

Step 2: Install Dependencies: The notebook installs required packages (`langfuse`, `langgraph`, `langchain`, `langchain-community`, `groq`, `requests`, `duckduckgo-search`) automatically via **`!pip install`**.

Step 3: Set Up API Keys in Colab Secrets

- In Google Colab, go to the “Secrets” tab (lock icon on the left sidebar).
- Add the following secrets with your own values:
 - Name: `GROQ_API_KEY`, Value: Your Groq API key.
 - Name: `LANGFUSE_API_KEY`, Value: Your Langfuse secret key.
 - Name: `LANGFUSE_PUBLIC_KEY`, Value: Your Langfuse public key.
 - Name: `LANGFUSE_HOST`, Value: Your Langfuse host URL.
- Ensure all keys are set correctly to avoid runtime errors.

Step 4: Configure Organization, Project, and IDs

In the code, update the following fields with your own information:

- Organization Name: Replace "first-index" with your Langfuse organization name (found in your Langfuse dashboard under account settings).
- Project Name: Replace "Langfuse_demo" with your Langfuse project name (found in your Langfuse dashboard).
- User ID: Replace "demo-user-001" with a unique user ID for your demo.

- Region (LANGFUSE_HOST): Ensure the LANGFUSE_HOST matches the region of your Langfuse deployment (e.g., <https://us.cloud.langfuse.com> for US region, or your self-hosted URL).

Step 5: Run the Notebook

Execute the cell in Colab.

- When prompted, enter a research topic (e.g., “quantum computing”).
- The notebook will output the search results, analysis, and summary, and log the trace to Langfuse.

Agent Development

We have previously discussed building agents in detail:

- This demo uses LangGraph to define a multi-agent workflow with a Search Agent, Analyzer Agent, and Summarizer Agent.
- Each agent performs a specific task (searching, analyzing, summarizing), and their interactions are orchestrated via a state graph.
- Langfuse enhances this setup by providing observability into their execution.

```
Enter a research topic (e.g., AI advancements, quantum computing): quantum computing
Researching topic: quantum computing
Search Results: MIT physicists predict exotic form of matter with potential for quantum computing. New work suggests the ability to create fractionalized electrons known as non-Abelian anyons. Another exciting 2025 development is the debut of portable quantum computing devices. Traditionally, quantum computers are massive installations requiring ultra-cold refrigerators or vacuum. Read the latest about the development of quantum computers. Skip to main content ... 2025 - Whether bismuth is part of a class of materials highly suitable for quantum computing and spintronics. Analysis: After analyzing the search results for the topic 'quantum computing', the following key points and trends have been identified:

**Key Points:**

1. **Advancements in exotic matter**: Researchers at MIT have predicted the possibility of creating fractionalized electrons known as non-Abelian anyons without a magnetic field, which could lead to more efficient quantum computing.
2. **Portable quantum computing devices**: A startup called SaxonQ has developed a compact quantum computer that can operate without the need for large, ultra-cold refrigerators or vacuum.
3. **New materials for quantum computing**: Bismuth is being explored as a potential material for quantum computing and spintronics, indicating ongoing research into finding suitable materials.

**Trends:**

1. **Increased focus on practical applications**: The development of portable quantum computing devices and the exploration of new materials suggest a growing emphasis on making quantum computing more accessible and usable in real-world scenarios.
2. **Rapid innovation**: The pace of discovery and development in quantum computing is accelerating, with new breakthroughs being reported frequently.

Research Summary: Quantum computing research highlights advancements in exotic matter, portable devices, and new materials. Trends show a focus on practical applications, making quantum computing more viable for commercial use.
Go to the Langfuse dashboard to view the trace:
URL: https://cloud.langfuse.com
Organization: first-index
Project: langfuse_demo
Look under the 'Traces' tab for 'multi-agent-research-langgraph' and check 'Sessions' for session tracking.
```

AI Observability with Langfuse

Langfuse is integrated to provide observability into the multi-agent system, enabling tracing, monitoring, and evaluation. Here's how it's implemented:

Step: Langfuse Initialization

A CallbackHandler is created to enable tracing of the LangGraph workflow

```
1 langfuse_handler = CallbackHandler(  
2     secret_key=LANGFUSE_SECRET_KEY,  
3     public_key=LANGFUSE_PUBLIC_KEY,  
4     host=LANGFUSE_HOST  
5 )
```

A Langfuse client is initialized for manual scoring:

```
1 langfuse = Langfuse(  
2     public_key=LANGFUSE_PUBLIC_KEY,  
3     secret_key=LANGFUSE_SECRET_KEY,  
4     host=LANGFUSE_HOST  
5 )
```

These use the API keys and host URL set in Colab Secrets, which have been configured with unique values.

Step: Tracing Agent Execution

- Each agent function (search_agent, analyzer_agent, summarizer_agent) is decorated with @observe() or @observe(as_type="generation") to enable Langfuse tracing.
- Within each agent, langfuse_context.update_current_observation logs metadata, inputs, and usage details:

```
1 langfuse_context.update_current_observation(  
2     input=f"User input: {user_input}\nSearch results: {search_results}",  
3     model="llama-3.3-70b-versatile",  
4     metadata={"organization": "first-index", "project": "Langfuse_demo", "agent": "analyzer_agent"}  
5 )
```

```
1 langfuse_context.update_current_observation(  
2     usage_details={  
3         "input": response.usage.prompt_tokens,  
4         "output": response.usage.completion_tokens  
5     }  
6 )
```

The metadata includes your organization and project names, which you've updated to match your Langfuse setup.

Step: Workflow Tracing

The LangGraph workflow is compiled with the Langfuse callback handler

```
1 app = workflow.compile()  
2 app = app.with_config({"callbacks": [langfuse_handler]})
```

This ensures all agent interactions are traced as a single workflow in Langfuse.

Step: Creating a Trace

After the workflow completes, a trace is created in Langfuse. The user_id, organization, and project fields should match your Langfuse configuration.

```
1 trace = langfuse.trace(  
2     name="multi-agent-research-langgraph",  
3     user_id="demo-user-001",  
4     session_id=session_id,  
5     metadata={  
6         "organization": "first-index",  
7         "project": "Langfuse_demo",  
8         "input_tokens": len(user_input.split()),  
9         "output_tokens": len(result["summary"].split())  
10    }  
11 )
```

Step: Scoring and Evaluation

- The trace is scored with a static relevance score and comment
- This score appears in the Langfuse dashboard under the trace, helping you evaluate the quality of the output.

```
1 trace.score(  
2     name="response-quality",  
3     value=0.9,  
4     comment="The research summary was concise and informative."  
5 )
```

Step: Flushing Events

`langfuse.flush()` ensures all tracing events are sent to the Langfuse server

Step: Viewing Traces

- The notebook provides instructions to view the trace in the Langfuse dashboard
- Update the URL, organization, and project names to match your Langfuse setup.

```
1 print("Go to the Langfuse dashboard to view the trace:")  
2 print("URL: https://cloud.langfuse.com")  
3 print("Organization: first-index")  
4 print("Project: Langfuse_demo")  
5 print("Look under the 'Traces' tab for 'multi-agent-research-langgraph' and check 'Sessions' for session tracking.")
```

Langfuse Interface

Click on the URL to look through the project insights

```

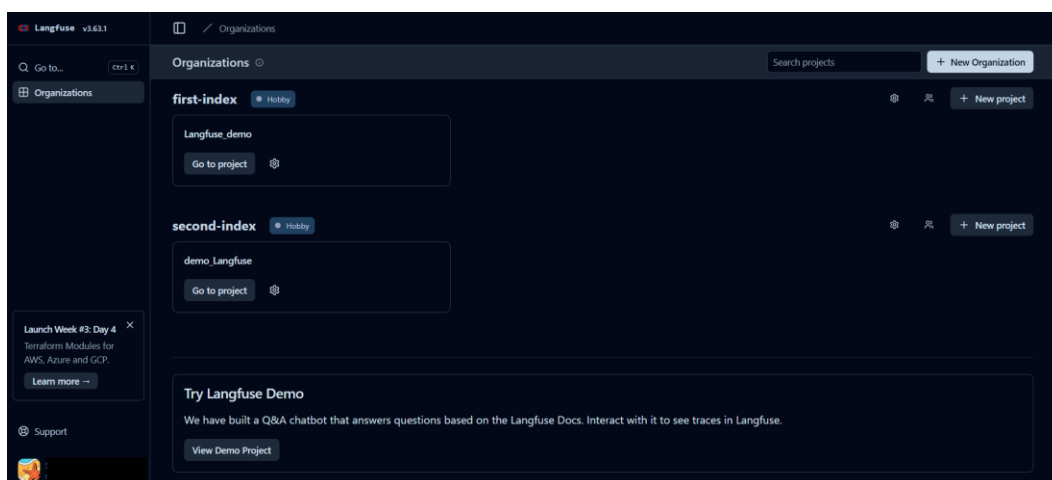
**Trends:**

1. **Increased focus on practical applications**
2. **R

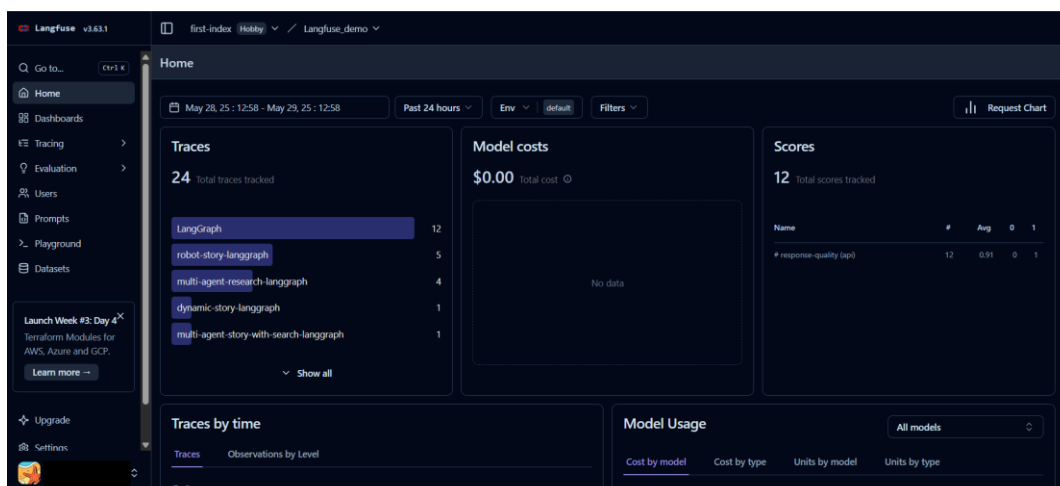
Research Summary: Quantum computing research high
Go to the Langfuse dashboard to view the trace:
URL: https://cloud.langfuse.com
Organization: first-index
Project: Langfuse_demo
Look under the 'Traces' tab for 'multi-agent-rese

```

You will get redirected here, to the organization's page. Click on the “Go to project” button under the designated organization for the project.



Project Home



Got to “Settings”

We go to the “LLM Connections” option and add a new API Key and model and save it.

Add new LLM API key

Provider name
Name to identify the key within Langfuse.
GEMINI_API_KEY

LLM adapter
Schema that is accepted at that provider endpoint.
google-ai-studio

API Base URL
Leave blank to use the default base URL for the given LLM adapter.
default

API Key
Your API keys are stored encrypted on our servers.
[Redacted]

Enable default models
Default models for the selected adapter will be available in Langfuse features. ☒

Custom models
Custom model names accepted by given endpoint.
gemini-2.0-flash

+ Add custom model name

Save new LLM API key

Got to “Scores/Evaluation”

Add new score configuration



Add new score config

Name
Agent Score

Data type
NUMERIC

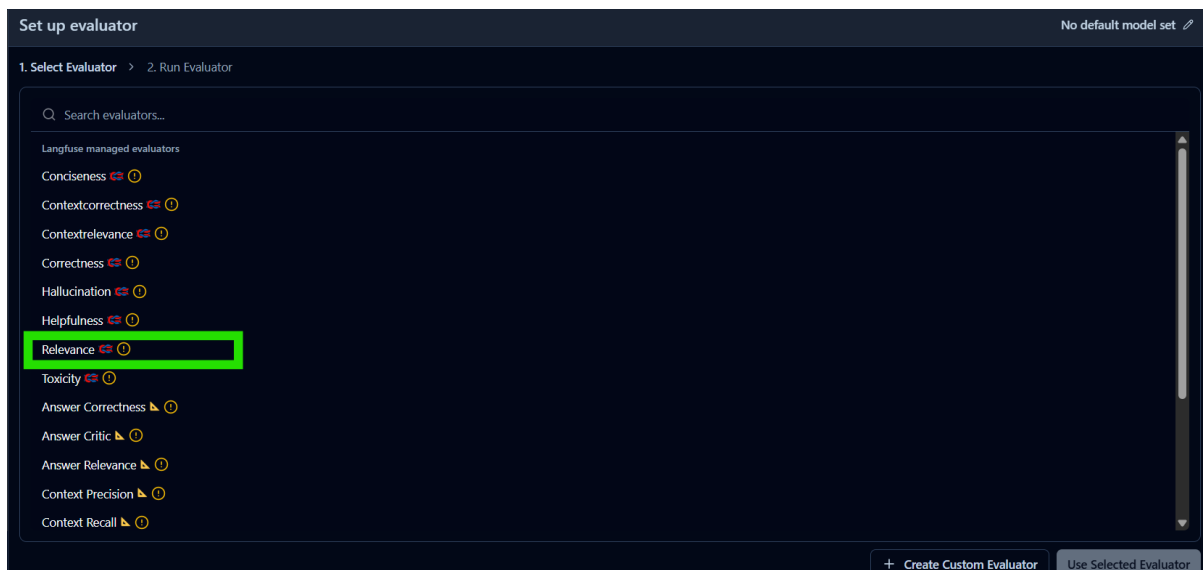
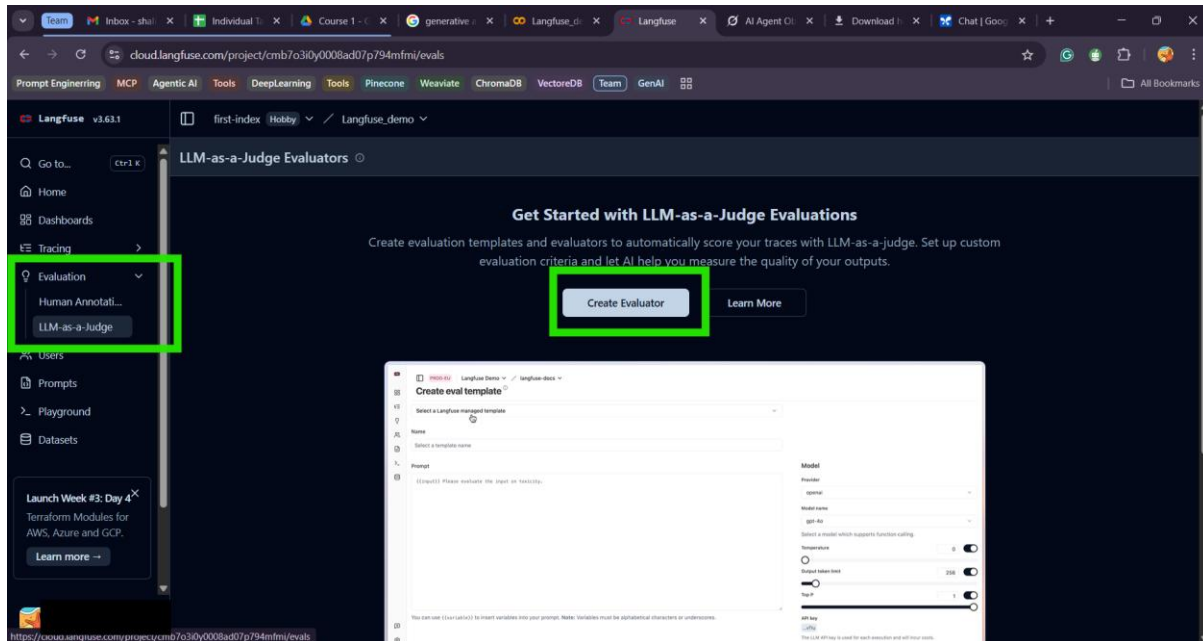
Minimum (optional)
0

Maximum (optional)
1

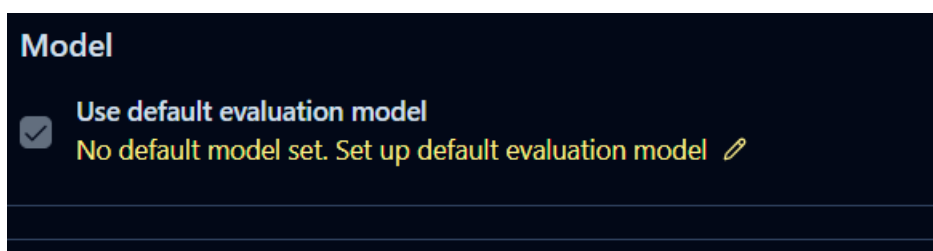
Description (optional)
Evaluating Relevance of the Model's response

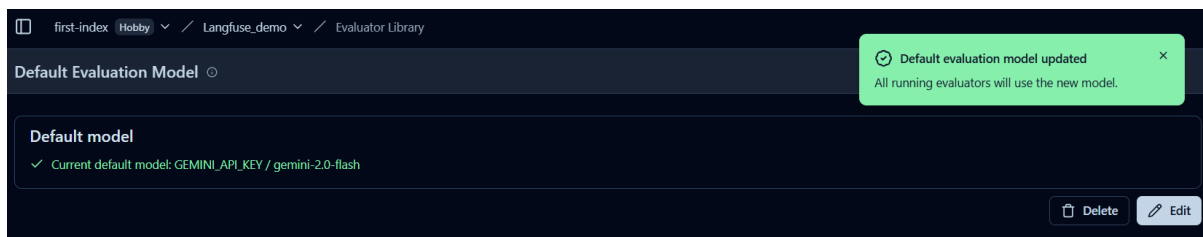
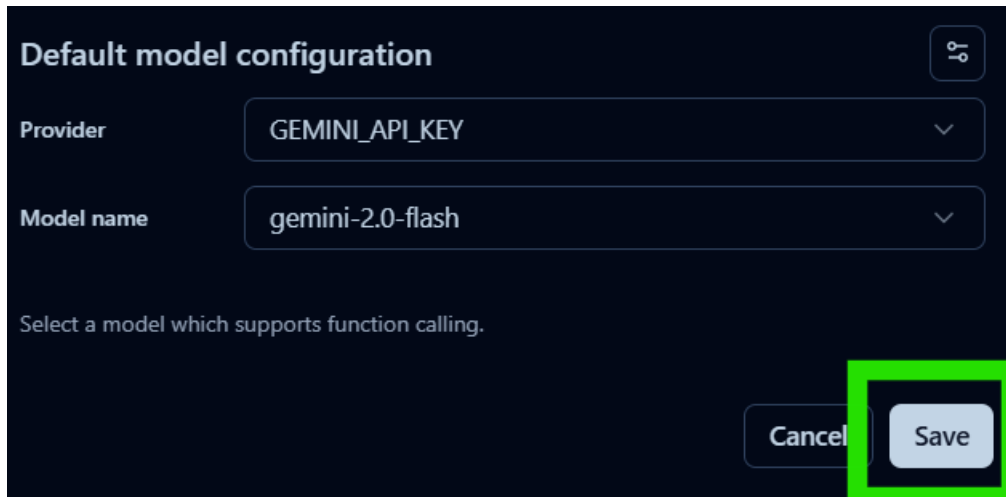
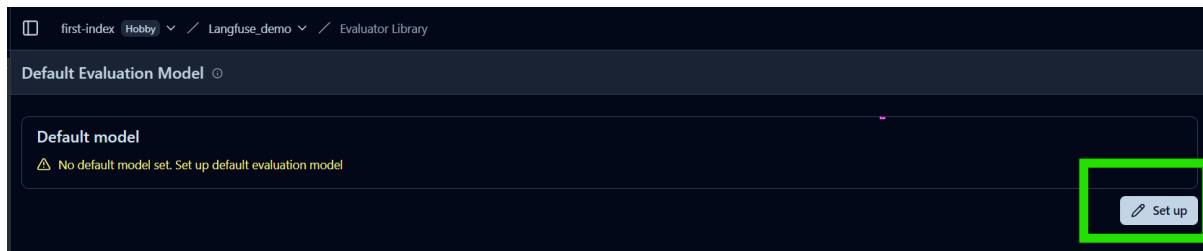
Submit

Creating Evaluator

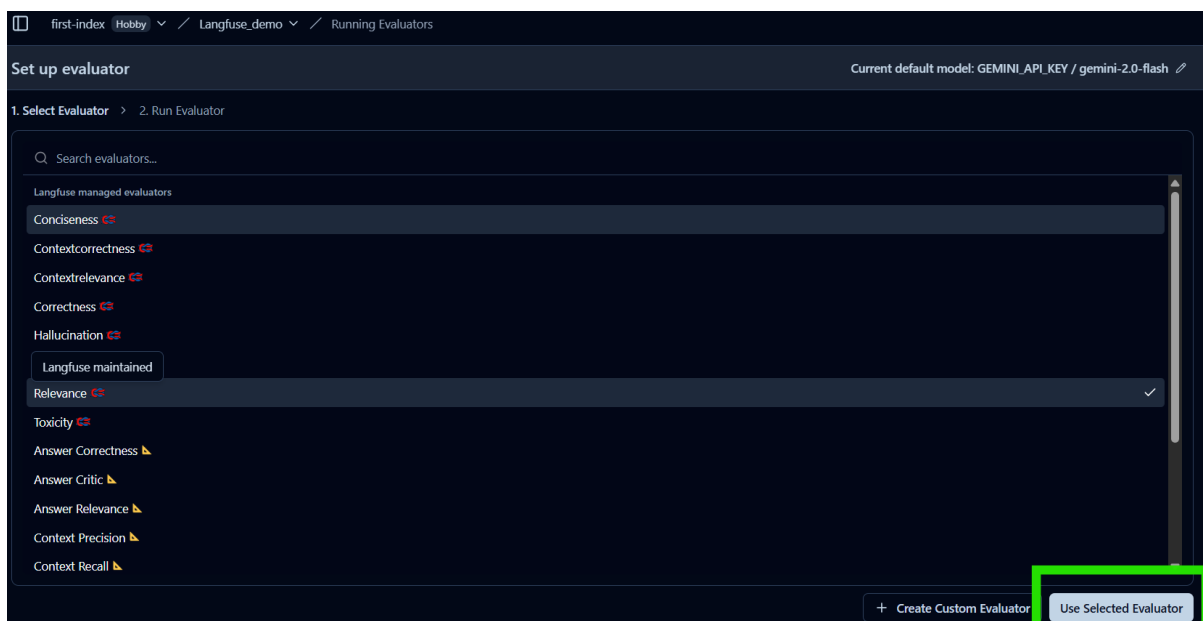


Setting a Default Model





Use Selected Evaluator



first-index Hobby / Langfuse_demo / Running Evaluators

Set up evaluator

Current default model: GEMINI_API_KEY / gemini-2.0-flash

1. Select Evaluator ✓ > 2. Run Evaluator: Relevance

Generated Score Name

Relevance

Target

Evaluator runs on

☒ New traces

☒ Existing traces ⓘ

Target data

Target filter

Preview sample matched traces

Sample over the last 24 hours that match these filters

Timestamp ▼	Name	Input	Output	Observation Level
2025-05-29 12:54:39	multi-agent-research-lang...			
2025-05-29 12:54:35	LangGraph	["user_input":"Vibe Coding"]	["user_input":"Vibe Coding","search_results":"Key Innovations in Vibe Cod...	4

first-index Hobby / Langfuse_demo / Running Evaluators

Set up evaluator

Current default model: GEMINI_API_KEY / gemini-2.0-flash

1. Select Evaluator ✓ > 2. Run Evaluator: Relevance

Variable mapping

Preview of the evaluation prompt with the variables replaced with the first matched trace data subject to the filters.

Evaluation Prompt Preview ⓘ

Evaluate the relevance of the generation on a continuous scale from 0 to 1. A generation can be considered relevant (Score: 1) if it enhances or clarifies the response, adding value to the user's comprehension of the topic in question. Relevance is determined by the extent to which the provided information addresses the specific question asked, staying focused on the subject without straying into unrelated areas or providing extraneous details.

Example:

Query: Can eating carrots improve your vision?

Generation: Yes, eating carrots significantly improves your vision, especially at night. This is why people who eat lots of carrots never need glasses. Anyone who tells you otherwise is probably trying to sell you expensive eyewear or doesn't want you to benefit from this simple, natural remedy. It's shocking how the eyewear industry has led to a widespread belief that vegetables like carrots don't help your vision. People are so gullible to fall for these money-making schemes.

Score: 0.1

Reasoning: Only the first part of the first sentence clearly answers the question and thus, is relevant. The rest of the text is not relevant to answer the query.

Input:

Query:

Generation:

Think step by step.

Variable mapping

{{query}} ⓘ

Object ⓘ

Object Variable ⓘ

JsonPath ⓘ

{{generation}} ⓘ

Object ⓘ

Object Variable ⓘ

JsonPath ⓘ

Active Evaluator after Execution

first-index Hobby / Langfuse_demo

LLM-as-a-Judge Evaluators ⓘ

Current default model: GEMINI_API_KEY / gemini-2.0-flash

Running Evaluators **Evaluator Library**

Generated Score Name	Status	Result	Logs	Referenced Evaluator	Created At ▼	Updated At	Target	Filter
Relevance	● active		View	Relevance	5/29/2025, 1:23:10 PM	5/29/2025, 1:23:10 PM	trace	

Evaluator Log View

Evaluator Relevance: 0fc6d50d-9ef8-496c-bae8-6c700b0d2f0d									
Filters Columns 9/9									
Status	Start Time	End Time	Score Name	Score Value	Score Comment	Error	Trace	Template	
pending	5/29/2025, 1:23:40 PM						c4f03cfe-8764-4f54-8a...	cma16wart005lynrdt	
pending	5/29/2025, 1:23:40 PM						fd10dca8-26ba-45e1-b...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	0	The generation is an er...		a8534215-db00-4832-...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	0	The query is empty. Th...		d27731ad-d8e2-4b1e-8...	cma16wart005lynrdt	
pending	5/29/2025, 1:23:40 PM						2534972e-c01e-40b9-a...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	1	The generation tells a s...		a366684b-f3e6-46df-a...	cma16wart005lynrdt	
pending	5/29/2025, 1:23:40 PM						6dcdd70b-c018-4841-a...	cma16wart005lynrdt	
pending	5/29/2025, 1:23:40 PM						283d9beb-1c6e-488d-...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	0	The query is empty. Th...		74930a3a-9f4e-48a2-a...	cma16wart005lynrdt	
pending	5/29/2025, 1:23:40 PM						9030a19b-29d4-4aa2-...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	0.9000	The generation tells a s...		289daa5e-bbcf-470a-a...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:12 PM	Relevance	0.9500	The generation provide...		9397fc58-a81f-49ab-81...	cma16wart005lynrdt	
completed	5/29/2025, 1:23:40 PM	5/29/2025, 1:24:11 PM	Relevance	0	The query is empty. Th...		d2710476-e911-4967-a...	cma16wart005lynrdt	

Traces:

first-index Hobby / Langfuse_demo					
Traces					
Search... Metadata Past 24 hours Filters Env default Table View 0 Columns 15/27					
	Timestamp	Name	Input	Output	Observation Levels
	2025-05-29 12:54:39	multi-agent-research-lang...			
	2025-05-29 12:54:35	LangGraph	["user_input": "Vibe Coding"]	["user_input": "Vibe Coding"; "search_results": "Key Innovations in Vibe Codi...	4
	2025-05-28 23:10:54	multi-agent-research-lang...			
	2025-05-28 23:10:48	LangGraph	["user_input": "quantum computing"]	["user_input": "quantum computing"; "search_results": "MIT physicists predic...	5
	2025-05-28 23:04:56	multi-agent-research-lang...			
	2025-05-28 23:04:53	LangGraph	["user_input": "quantum computing"]	["user_input": "quantum computing"; "search_results": "MIT physicists predic...	4
	2025-05-28 22:53:51	multi-agent-research-lang...			
	2025-05-28 22:53:48	LangGraph	["user_input": "quantum computing"]	["user_input": "quantum computing"; "search_results": "MIT physicists predic...	4
	2025-05-28 22:51:38	multi-agent-story-with-sea...			
	2025-05-28 22:51:35	LangGraph	["user_input": "AI"]	["user_input": "AI"; "search_results": "history of artificial intelligence (AI), a su...	5
	2025-05-28 22:45:48	multi-agent-story-langgraph			
	2025-05-28 22:45:46	LangGraph	["user_input": "dragon"]	["user_input": "dragon"; "story_idea": "As the last dragon on earth lay dying, ...	4
	2025-05-28 22:41:15	dynamic-story-langgraph			
	2025-05-28 22:41:14	LangGraph	["user_input": "AI"]	["user_input": "AI"; "story": "In the year 2157, in a world where artificial intelli...	2

Sessions:

first-index Hobby / Langfuse_demo

Sessions

Past 24 hours Filters Env default Table View 0 Columns 11/17

ID	Created At	Duration	Environment	# Relevance (eval)	# response-quality (a...	User IDs	Traces	Total C
ee3b46fd-6c79-47b2-97c0-c3d8...	2025-05-29 12:54:39		default			demo-user-001	1	\$0.00
d5f9bec9-dc34-4dfe-8414-f377f...	2025-05-28 23:10:54		default			demo-user-001	1	\$0.00
523bef05-371a-479c-89fb-48113...	2025-05-28 23:04:56		default			demo-user-001	1	\$0.00
f2656ebd-fd4c-4843-b32d-26d8...	2025-05-28 22:53:51		default			demo-user-001	1	\$0.00
0e51e46c-87d1-451f-a798-8c661...	2025-05-28 22:51:38		default			demo-user-001	1	\$0.00
39517a50-3188-4a45-bc2c-9741...	2025-05-28 22:45:48		default			demo-user-001	1	\$0.00
f09b4894-0c74-4d0d-97cd-2f8d...	2025-05-28 22:37:08		default			demo-user-001	2	\$0.00
9e835d7a-424b-436a-ba49-54b...	2025-05-28 22:26:52		default			demo-user-001	1	\$0.00
14769f89-eea1-4989-9866-19c5e...	2025-05-28 22:24:16		default			demo-user-001	1	\$0.00
76733f69-7822-4c55-a671-0b184...	2025-05-28 22:21:50		default			demo-user-001	1	\$0.00
63a729f2-b82e-4bf1-a791-afd9...	2025-05-28 22:16:59		default			demo-user-001	1	\$0.00

Scores:

first-index Hobby / Langfuse_demo

Scores

Past 24 hours Filters Env default Table View 0 Columns 16/18

Trace Name	Trace	Observation	Session	Environment	User	Timestamp	Source	Name	Data Type	Value
robot-story-langgraph	283d9beb-1c...			default	demo-user-0...	2025-05-29 13:26:13	EVAL	Relevance	NUMERIC	0
LangGraph	9397fc58-a8...			default		2025-05-29 13:24:12	EVAL	Relevance	NUMERIC	0.9500
multi-agent-research-l...	eb5ed48b-f0...			default	demo-user-0...	2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0
LangGraph	04e894da-cl...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	0192b625-87...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	21fc4059-76...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	a366804b-f3...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	a8534215-db...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0
LangGraph	e9c7b7ac-20...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	ab53a118-d0...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	1
LangGraph	289daa5e-b...			default		2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0.9000
robot-story-langgraph	74930a3a-9f...			default	demo-user-0...	2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0
multi-agent-story-lang...	d2710476-e9...			default	demo-user-0...	2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0
Automatic chain-termina...	96a073cc-46...			default	demo-user-0...	2025-05-29 13:24:11	EVAL	Relevance	NUMERIC	0

Dashboards

Langfuse v3.63.1

first-index Hobby / Langfuse_demo

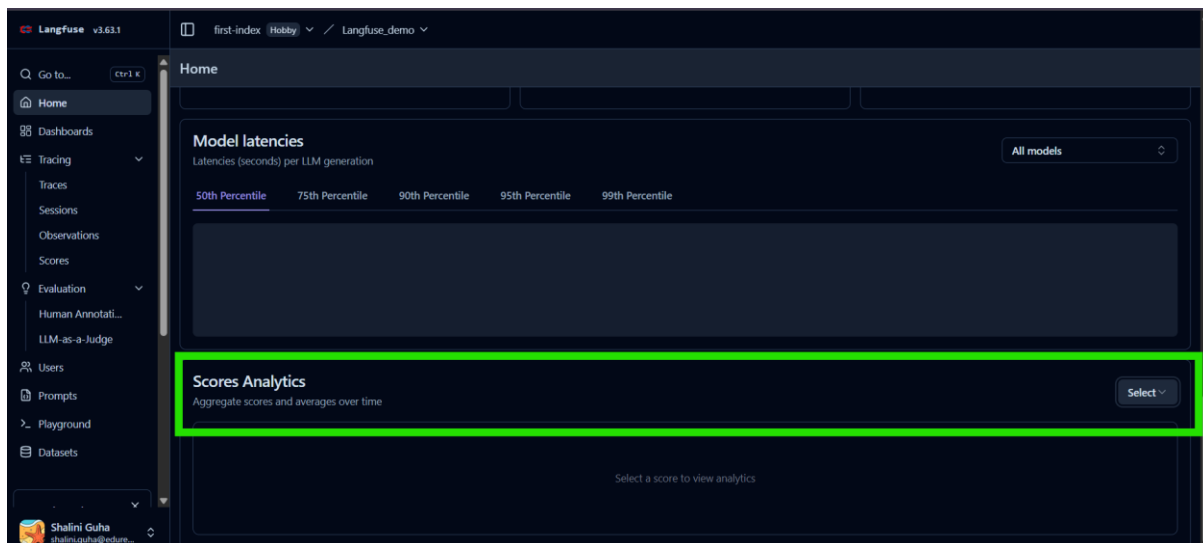
Dashboards

+ New dashboard

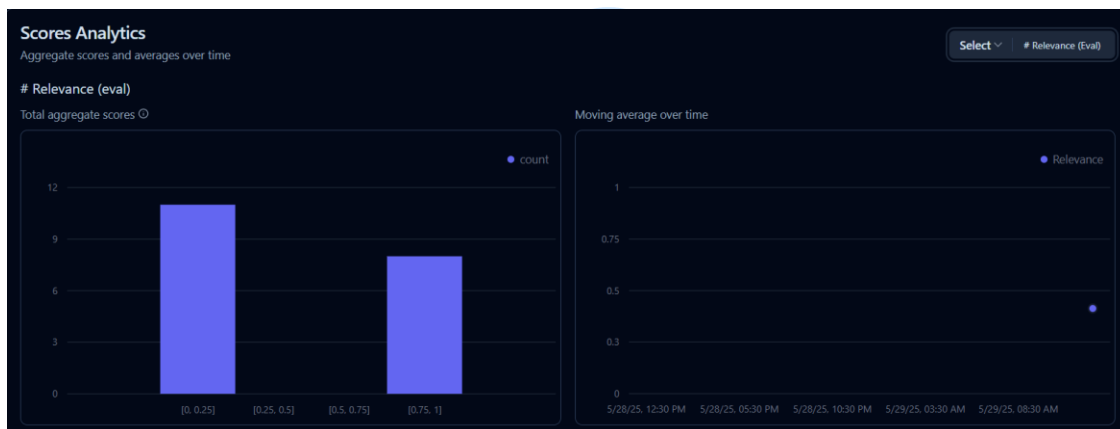
Name	Description	Owner	Created At	Updated At	Actions
Langfuse Cost Dashboard	Review your LLM costs.	Langfuse	2025-05-20 21:08:32	2025-05-20 21:39:56	
Langfuse Usage Management	Track usage metrics across traces, observations, and scores to manage resource allocation.	Langfuse	2025-05-20 19:48:27	2025-05-20 21:26:46	
Langfuse Latency Dashboard	Monitor latency metrics across traces and generations for performance optimization.	Langfuse	2025-05-20 19:06:15	2025-05-20 21:26:46	

Project Home

Scroll to the Score Analytics and select the desired analytic to view.



Relevance Evaluation



Response Quality

